nf-core / **scrnaseq** Public

Code Issues 62 Pull requests 12 Actions Security and quality Insights

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nictru Merge pull request #485 from nf-core/dev f7bf36d · 7 months ago

.devcontainer	Template update for nf-core/...	7 months ago
.github	Attempt to fix awsfulltest	7 months ago
assets	Merge remote-tracking bran...	7 months ago
bin	correct parsed ocm barcode...	10 months ago
conf	Merge branch 'dev' into nf-c...	10 months ago
docs	Merge remote-tracking bran...	7 months ago
lib	Merge remote-tracking bran...	2 years ago
modules	Merge remote-tracking bran...	7 months ago
subworkflows	Merge remote-tracking bran...	7 months ago
tests	Update test snapshots	7 months ago
workflows	Merge pull request #483 fro...	10 months ago
.gitattributes	Template update for nf-core/...	4 years ago
.gitignore	Merge branch 'dev' into nf-c...	2 years ago
.nf-core.yml	Merge remote-tracking bran...	7 months ago
.pre-commit-config.ya...	Template update for nf-core/...	7 months ago
.prettierignore	Template update for nf-core/...	7 months ago
.prettierrc.yml	Template update for nf-core/...	last year
.ruff.toml	Don't format modules and su...	last year
CHANGELOG.md	Update CHANGELOG	9 months ago

CITATIONS.md	Merge branch 'dev' into nf-c...	2 years ago
CODE_OF_CONDUCT....	Template update for nf-core/...	3 years ago
LICENSE	Template update for nf-core/...	2 years ago
README.md	Merge remote-tracking bran...	7 months ago
main.nf	Merge remote-tracking bran...	7 months ago
modules.json	Merge remote-tracking bran...	7 months ago
nextflow.config	Merge remote-tracking bran...	7 months ago
nextflow_schema.json	Merge remote-tracking bran...	7 months ago
nf-test.config	change nf-test function so th...	9 months ago
ro-crate-metadata.json	Fix ro-crate	7 months ago
tower.yml	Turn off filtered gtf publishin...	4 years ago



Open in GitHub Codespaces | Run nf-test passing | nf-core linting passing | CI tests full size

DOI 10.5281/zenodo.3568187 | unit tests nf-test

version ≥25.04.0 | nf-core template 3.4.1 | run with conda | run with docker | run with singularity

Launch Seqera Platform

slack nf-core #scrnaseq | bluesky @nf_core | mastodon nf_core | youtube nf-core

Introduction

nf-core/scrnaseq is a bioinformatics best-practice analysis pipeline for processing 10x Genomics single-cell RNA-seq data.

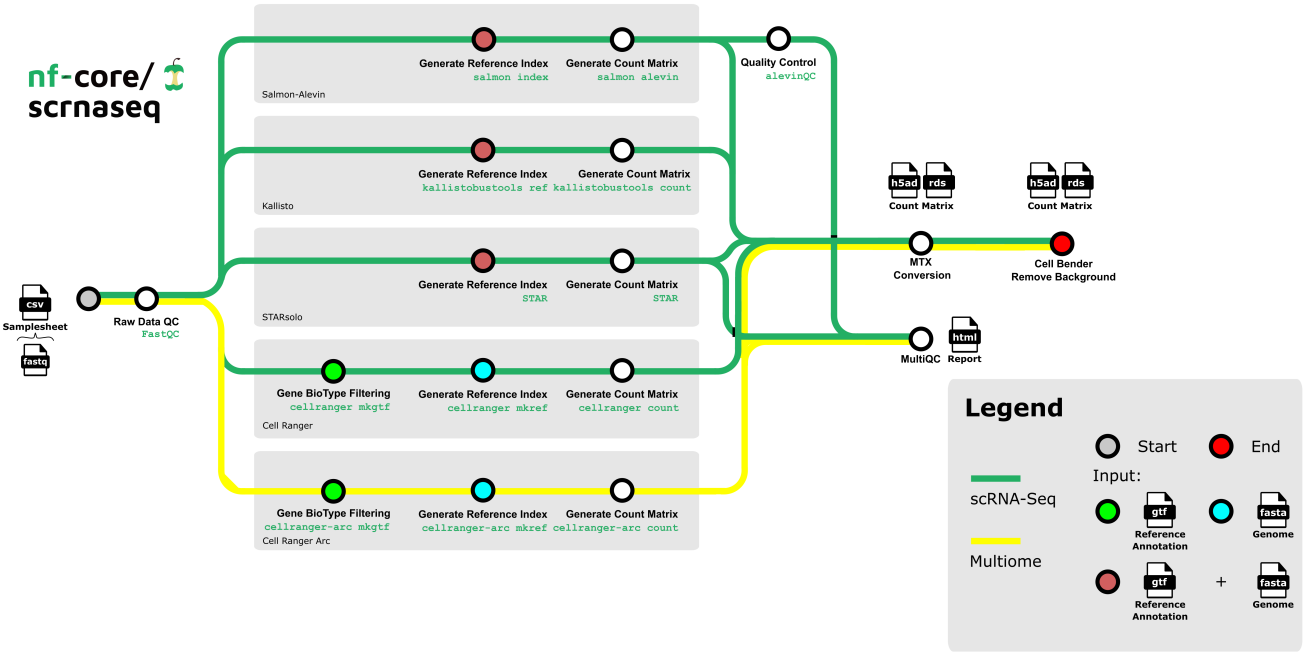
This is a community effort in building a pipeline capable to support:

- SimpleAF(Alevin-Fry) + AlevinQC
- STARSolo
- Kallisto + BUStools
- Cellranger

> **Important**
>
> Cellranger is a commercial tool from 10X Genomics Inc. and falls under the EULA from 10X Genomics Inc. The container provided for the CellRanger functionality in this pipeline has been built by the nf-core community and is therefore *not supported by 10X genomics* directly. We are in discussions with 10X on how to improve the user experience and licence situation for both us as a community as well as 10X and end users and will update this statement here accordingly.

Documentation

The nf-core/scrnaseq pipeline comes with documentation about the pipeline usage, parameters and output.



Usage

> **Note**
>
> If you are new to Nextflow and nf-core, please refer to this page on how to set-up Nextflow. Make sure to test your setup with `-profile test` before running the workflow on actual data.

First, prepare a samplesheet with your input data that looks as follows:

`samplesheet.csv` :

```
sample,fastq_1,fastq_2,expected_cells
pbmc8k,pbmc8k_S1_L007_R1_001.fastq.gz,pbmc8k_S1_L007_R2_001.fastq.gz,10000
pbmc8k,pbmc8k_S1_L008_R1_001.fastq.gz,pbmc8k_S1_L008_R2_001.fastq.gz,10000
```

Each row represents a fastq file (single-end) or a pair of fastq files (paired end).

Now, you can run the pipeline using:

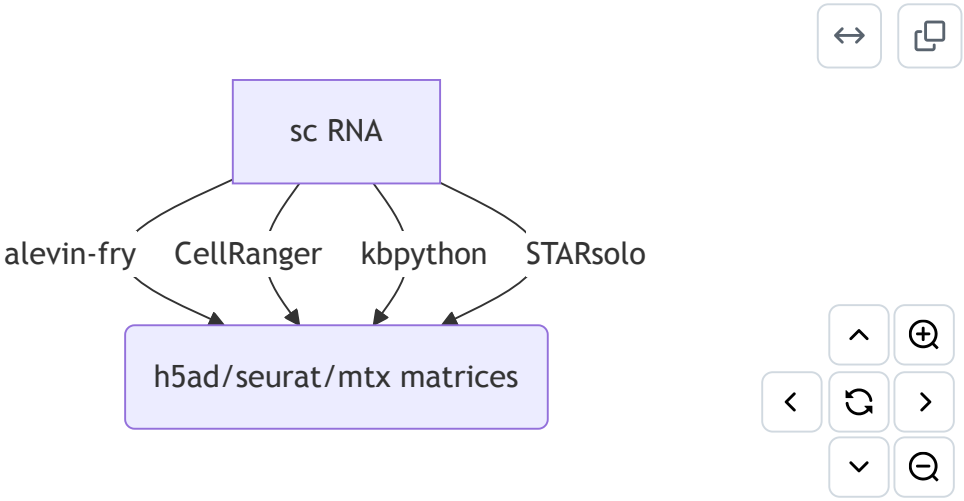
```
nextflow run nf-core/scrnaseq \
   -profile <docker/singularity/.../institute> \
   --input samplesheet.csv \
   --fasta GRCm38.p6.genome.chr19.fa \
   --gtf gencode.vM19.annotation.chr19.gtf \
   --protocol 10XV2 \
   --aligner <simpleaf/kallisto/star/cellranger> \
   --outdir <OUTDIR>
```

> **Warning**
>
> Please provide pipeline parameters via the CLI or Nextflow `-params-file` option. Custom config files including those provided by the `-c` Nextflow option can be used to provide any configuration ***except for parameters***; see docs.

For more details and further functionality, please refer to the usage documentation and the parameter documentation.

Decision Tree for users

The nf-core/scrnaseq pipeline features several paths to analyze your single cell data. Future additions will also be done soon, e.g. the addition of multi-ome analysis types. To aid users in analyzing their data, we have added a decision tree to help people decide on what type of analysis they want to run and how to choose appropriate parameters for that.



Options for the respective alignment method can be found here to choose between methods.

Pipeline output

To see the results of an example test run with a full size dataset refer to the results tab on the nf-core website pipeline page. For more details about the output files and reports, please refer to the output documentation.

Credits

nf-core/scrnaseq was originally written by Bailey PJ, Botvinnik O, Marques de Almeida F, Gabernet G, Peltzer A, Sturm G.

We thank the following people and teams for their extensive assistance in the development of this pipeline:

- @heylf
- @KevinMenden
- @FloWuenne
- @rob-p
- GHGA

Contributions and Support

If you would like to contribute to this pipeline, please see the contributing guidelines.

For further information or help, don't hesitate to get in touch on the Slack `#scrnaseq` channel (you can join with this invite).

Citations

If you use nf-core/scrnaseq for your analysis, please cite it using the following doi: 10.5281/zenodo.3568187

The basic benchmarks that were used as motivation for incorporating the available modular workflows can be found in this publication.

We offer all three paths for the processing of scRNAseq data so it remains up to the user to decide which pipeline workflow is chosen for a particular analysis question.

An extensive list of references for the tools used by the pipeline can be found in the `CITATIONS.md` file.

Releases 18

4.1.0 Latest
on Oct 30, 2025

+ 17 releases

Contributors 49



+ 35 contributors

Languages

Nextflow 81.3% Python 14.9% R 1.8% HTML 1.3% Other 0.7%